Modelling & Simulation: Final Assignment

Integration: Robot arm with actuation and control

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Introduction

It is time to integrate everything that you have learned. So far you have seen electric motors, their control, series-elastic actuators and torque control, energy-based concepts, 2-D robot arms, 3-D rigid body dynamics, robot arms in 3-D, and impedance control. For this final assignment you will have double the time, and you will both 1) write a report on it, and 2) be able to defend it in your oral exam.

Assignment

Model a 3-D, 3-DoF robot arm with rotational joints. This may be done using explicit modelling of the dynamics or using 20-sim's 3D Mechanics Editor. Include the full actuator dynamics and (cascaded) control; series-elastic actuation with torque and current control (discrete-time digital control). The arm should perform a motion task in 3-D space, use Cartesian impedance control, and be gravity compensated. Besides integrating your existing knowledge, focus on getting realistic parameters and verification of your model and simulation results.

Questions you should answer in the process:

- In your report, clearly explain the formulation of your control (current, torque, impedance, gravity comp.).
- How do you choose parameters for each of the components and constitutive elements? Can you use existing datasheets? Make sure these are realistic and well-motivated.
- Verify your model and simulation and show this in the report. Can you isolate parts of your model to verify them separately? Are the currents realistic? Torques? Speeds?
- What impedance parameters will you choose for your Cartesian impedance control and why? Can you show the difference? Make sure you include at least one experiment where the robot interacts with its environment to show this.
- Use animations – Can you show some images or even link to videos?

A few hints:

- As always, consider simplifying initially, and perform **explicit verification steps**. For example,
  1. Leave the arm unactuated to see if it swings under gravity as you would expect,
  2. Develop and test your actuators separately, with fixed output or a simple mass as load (this also allows to parallelise the work within the group!),
  3. Try to get your impedance control working first by directly applying torques instead of using your actuator models,

4. This is simulation; you can always turn gravity off during testing.
5. Try to break your own model; what happens when you change parameters?